

Detailed Performance Report: Task 4 - Classification Models

1. Project Overview

Following the guidelines in **Artificial Intelligence & Machine Learning - Task 4**, we implemented a binary classification pipeline using the Breast Cancer Wisconsin dataset. The goal was to identify samples as either Malignant (0) or Benign (1).

2. Methodology

- **Data Split:** 80% Training, 20% Testing.
- **Models Evaluated:**
 - Baseline Logistic Regression
 - Decision Tree Classifier
 - Weighted Logistic Regression (to address potential class imbalance)

3. Comparative Analysis of Results

Model	Accuracy	Precision (Malignant)	Recall (Malignant)	F1-Score (Malignant)
Logistic Regression	96%	0.97	0.91	0.94
Decision Tree	95%	0.93	0.93	0.93
Weighted LogReg	97%	0.98	0.95	0.96

4. Key Findings & Metric Interpretation

- **Confusion Matrix Analysis:** The standard Logistic Regression model missed 4 malignant cases (False Negatives). In a medical context, Recall is critical because missing a malignant diagnosis is more costly than a false alarm.
- **Class Weighting:** By applying `class_weight='balanced'`, we improved the model's sensitivity. The Recall for the malignant class rose from **0.91 to 0.95**, successfully identifying more true positive cases.
- **ROC Curve & AUC:** The ROC curve area of **1.00** suggests that the model has perfect separability potential on this test set. This confirms that the Logistic Regression model is highly robust for this specific feature set.
- **Decision Tree vs. Linear Models:** The Decision Tree showed slightly lower precision. Given the relatively high dimensionality (30 features), the linear boundary of Logistic Regression proved more effective at generalizing than the axis-parallel splits of the Decision Tree.

5. Conclusion

The **Weighted Logistic Regression** is the recommended model for this task. It provides the best balance between overall accuracy and the critical recall of the malignant class, fulfilling the 'Core Industry Skill' of handling imbalanced data requirements outlined in the PDF.